CHM2050 Practice Exam A (2nd Mid-Term style) Name: _____

Check Team: □ Argon, □ Barium, □ Cobalt, □ Dysprosium, □ Europium, □ Fluorine, □ Gold.

Answer all questions neatly in the spaces provided. Equations, conversion factors, natural constants, an electronegativity table, and a periodic table are provided in the potentially helpful information at the back of this exam. **Give numerical answers with units and the correct number of significant digits for full credit. Show your work or there will be no points.** The exam consists of sixteen problems with a nominal total of 225 points; it is counted out of 200. Suggested time for each problem is listed. Read every question first and start with the ones you find easy. *Target time: 110 min.*

Problem 1 (15 points, 10 min):

Draw Lewis diagrams, using the Natta projection (line-dash-wedge notation) if necessary, and predict the correct molecular shape of the following species based on VSEPR theory. Tell whether these compounds are polar or non-polar. If polar, draw the molecular dipole moment vector using the IUPAC convention.

(a) Chlorine trifluoride, ClF_3. (b) Xenon difluoride, XeF_2. (c) Phosphorus trichloride, PCl_3.

Problem 2 (20 points, 10 min):

Describe the molecular bonds in the ethene molecule, aka ethylene, C_2H_4, using concepts from valence bond theory. For full credit: write the correct Lewis structure, give the local geometry, bond angles, and hybridization of the two carbon atoms. Sketch the principal axes (x, y, and z) of a 3D coordinate system next to your structure. For each bond in C_2H_4, indicate whether it is a σ- or a π-bond and which atomic and/or hybrid orbitals overlap to form it. Sketch the bond orbital(s) between the two carbon atoms. Give the total number of σ- and π-bonds in the molecule.

Problem 3 (20 points, 10 min):

Consider the nitrosonium cation, NO^+, the nitric oxide radical, $NO^\bullet$, and the nitroxyl anion, NO^-. Based on the molecular orbital diagram for 2nd row diatomic molecules (use the ordering valid for N_2), write the valence electron configuration of each species, give its bond order, tell whether it is para- or diamagnetic, and assign the correct bond length to each species. The bond lengths in the gas phase are 106.3 pm, 115.1 pm, and 125.8 pm. In which atomic orbitals does the electron density of the bonding MOs predominantly reside, N or O? Why?

Problem 4 (15 points, 10 min):

Methyl isocyanate, CH_3NCO, is a highly toxic industrial intermediate (Bhopal disaster, 1984). The atoms are connected in the order $H_3C–N–C–O$. Draw the correct Lewis structure including all formal charges that are not zero, give the hybridization of each C, N, and O atom, and estimate the C–N–C and N–C–O bond angles. Determine the molar mass. Show work for full credit.

Problem 5 (10 points, 5 min):

The black magnetic mineral magnetite is an iron oxide with the experimental %-composition 72.36% Fe and 27.64% O by mass. Determine its empirical formula. Show your work for full credit.

Problem 6 (15 points, 5 min):

Caffeine has the %-composition 49.48% C, 5.19% H, 28.85% N, and 16.48% O by mass. Mass spectrometry gives a molar mass of about 194 g/mol. Determine the empirical and the molecular formula of caffeine. Show your work for full credit.

Problem 7 (20 points, 10 min):

Iron is produced in a blast furnace by the reduction of iron(III) oxide with carbon monoxide according to the (unbalanced) equation:

$$\mathbf{Fe_2O_3(s) + CO(g) \rightarrow Fe(l) + CO_2(g)}$$

(a) Correctly balance this rxn equation using the smallest whole numbers as coefficients.
(b) 50.0 g Fe_2O_3 is reacted with 25.0 g CO. Determine the limiting reactant.
(c) Determine the theoretical yield of Fe and the mass of excess reactant left over.
(d) 28.9 g Fe was actually isolated. Determine the %-yield of the rxn.

Potentially helpful information: molar weights in g/mol: 159.687 (Fe_2O_3); 28.010 (CO); 55.845 (Fe); 44.009 (CO_2).

Problem 8 (10 points, 5 min):

Nigel needs 250. mL of 0.0750 M HCl(aq) for a titration. The stockroom only has 6.00 M HCl(aq). Calculate the volume of stock solution he needs and describe in two or three sentences how he should prepare the solution (name the glassware).

Problem 9 (15 points, 15 min):

Explain in words how to make a 0.2000 molal aqueous solution of copper(II) sulfate that contains a total of 500.0 g of water as solvent, using unlimited de-ionized water and a bottle of $CuSO_4 \cdot 5H_2O$ (copper(II) sulfate pentahydrate, “blue vitriol”). Be quantitative in your description. In one sentence, explain why molality is a more precise concentration measure than molarity.

Problem 10 (15 points, 10 min):

Describe the reaction that takes place when 100. mL of a 0.0200 M aqueous solution of nickel(II) chloride is mixed with 100. mL of a 0.0300 M aqueous solution of sodium hydroxide. For full credit (a) take qualitative chemical inventory of the two reactant solutions; (b) tell what type of reaction takes place; (c) write out the net ionic equation and name the spectator ions; (d) determine the limiting reactant and the mass of solid formed ($Ni(OH)_2$: 92.707 g/mol).

Problem 11 (15 points, 5 min):

A 25.00-mL sample of sulfuric acid of unknown concentration is titrated with 0.1500 M NaOH(aq). The phenolphthalein end point is reached after 31.42 mL of titrant has been added. Write the balanced molecular equation of the neutralization and determine the molarity of the sulfuric acid. What is the difference between the equivalence point and the end point?

Problem 12 (15 points, 10 min):

Zinc metal reacts with excess hydrochloric acid to give hydrogen gas: $Zn(s) + 2\ HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$. The H_2 is collected over water at 25.0 °C. The volume of collected gas is 245 mL and the total pressure is 755 mm-Hg. What mass of zinc reacted?

Potentially helpful information: vapor pressure of water at 25.0 °C = 23.8 mm-Hg; molar mass of Zn = 65.38 g/mol. Assume ideal gas behavior.

Problem 13 (15 points, 10 min):

A 500.-mL glass bulb is filled with 1.28 g of an unknown gaseous oxide of a non-metal at 27 °C and 0.987 atm. (a) Determine the molar mass of the gas and suggest its identity (it is one of the "SO_x" pollutants). (b) Determine its density at these conditions. (c) How fast does it effuse relative to $O_2(g)$? (d) Draw its Lewis structure and give its shape.

Problem 14 (10 points, 5 min):

You need to transfer 75 mL of a solution of *tert*-butyl lithium in pentane from a reagent bottle into a reaction flask. (a) Give two different and specific hazards this operation presents. (b) Which transfer technique should be used for this volume? (c) Give three pieces of PPE required. (d) Which section of the Safety Data Sheet would you consult for the required PPE?

Problem 15 (10 points, 5 min):

Copper metal dissolves in concentrated nitric acid with the evolution of a brown gas:

$$\mathbf{Cu(s) + 4\ HNO_3(aq) \rightarrow Cu(NO_3)_2(aq) + 2\ NO_2(g) + 2\ H_2O(l)}$$

Assign oxidation numbers to every element in all reactants and products. Identify the element that is oxidized, the element that is reduced, the oxidizing agent, and the reducing agent.

Problem 16 (10 points, 5 min):

(a) Rank He(g), N_2(g), and SO_2(g) at 25 °C by increasing average kinetic energy and by increasing root-mean-square speed. (b) Calculate the root-mean-square speed of N_2(g) at 25 °C. (c) Under which conditions (T and P) does a real gas deviate most strongly from ideal behavior, and why?

Potentially Helpful Information (use at your own risk since not all equations are applicable at all times)

Conversion Factors:

$$
\begin{aligned}
1\ \mathrm{kg} &= 2.20462\ \mathrm{lb} \\
1\ \mathrm{lb} &= 16\ \mathrm{oz} \\
1\ \mathrm{mile} &= 1.60934\ \mathrm{km} \\
1\ \mathrm{in} &= 2.54\ \mathrm{cm} \\
1\ \mathrm{ft} &= 0.3048\ \mathrm{m} \\
1\ \mathrm{m} &= 1.09361\ \mathrm{yd} \\
1\ \mathrm{dm^3} &= 1\ \mathrm{L} \\
1\ \mathrm{cm^3} &= 1\ \mathrm{mL} \\
1\ \mathrm{L} &= 1.05669\ \mathrm{qt} \\
1\ \mathrm{Pa} &= 1\ \frac{\mathrm{kg}}{\mathrm{m \cdot s^2}} \\
1\ \mathrm{atm} &= 760\ \mathrm{Torr} = 760\ \mathrm{mm\ Hg} \\
1\ \mathrm{atm} &= 1.01325 \cdot 10^5\ \mathrm{Pa} \\
1\ \mathrm{bar} &= 1 \cdot 10^5\ \mathrm{Pa} \\
1\ \mathrm{J} &= 1\ \frac{\mathrm{kg \cdot m^2}}{\mathrm{s^2}} \\
1\ \mathrm{J} &= 1\ \mathrm{C \cdot V} \\
1\ \mathrm{J} &= 1\ \mathrm{W \cdot s} \\
1\ \mathrm{cal} &= 4.184\ \mathrm{J} \\
1\ \mathrm{atm \cdot L} &= 1.01325 \cdot 10^2\ \mathrm{J} \\
1\ \mathrm{eV} &= 1.602176634 \cdot 10^{-19}\ \mathrm{J} \\
0\ ^\circ\mathrm{C} &= 273.15\ \mathrm{K} \\
1\ \mathrm{ppm} &= 1 \cdot 10^{-6} \\
1\ \text{Å} &= 1 \cdot 10^{-10}\ \mathrm{m} \\
1\ \mathrm{Ry} &= 13.605698\ \mathrm{eV} \\
1\ \mathrm{Da} &= 1\ \mathrm{u} = 1\ \mathrm{g/mol}
\end{aligned}
$$

Natural Constants (†indicates an exact number):

$$
\begin{aligned}
N_\mathrm{A}^\dagger &= 6.02214076 \cdot 10^{23}\ \mathrm{mol^{-1}} \\
e^\dagger &= 1.602176634 \cdot 10^{-19}\ \mathrm{C} \\
F^\dagger &= 9.648533212 \cdot 10^4\ \frac{\mathrm{C}}{\mathrm{mol}} \\
m_e &= 9.1093837015 \cdot 10^{-31}\ \mathrm{kg} \\
m_p &= 1.67262192369 \cdot 10^{-27}\ \mathrm{kg} \\
m_n &= 1.67492749804 \cdot 10^{-27}\ \mathrm{kg} \\
h^\dagger &= 6.62607015 \cdot 10^{-34}\ \mathrm{J \cdot s} \\
\hbar &= h/(2\pi) = 1.054571817 \cdot 10^{-34}\ \mathrm{J \cdot s} \\
c^\dagger &= 2.99792458 \cdot 10^8\ \frac{\mathrm{m}}{\mathrm{s}} \\
g^\dagger &= 9.80665\ \frac{\mathrm{m}}{\mathrm{s^2}} \\
R^\dagger &= 8.314462618\ \frac{\mathrm{J}}{\mathrm{K \cdot mol}} \\
R &= 8.205736 \cdot 10^{-2}\ \frac{\mathrm{L \cdot atm}}{\mathrm{K \cdot mol}} \\
k_\mathrm{B}^\dagger &= 1.380649 \cdot 10^{-23}\ \frac{\mathrm{J}}{\mathrm{K}} \\
\varepsilon_0 &= 8.8541878128 \cdot 10^{-12}\ \frac{\mathrm{As}}{\mathrm{Vm}} \\
R_\mathrm{H} &= 1.096775834 \cdot 10^7\ \mathrm{m^{-1}} \\
R_\infty &= 1.0973731568160 \cdot 10^7\ \mathrm{m^{-1}}
\end{aligned}
$$

Equations:

$$
\begin{aligned}
V_\mathrm{cube} &= \ell^3 \\
V_\mathrm{cylinder} &= \left(\pi r^2\right) \cdot \ell \\
V_\mathrm{sphere} &= \frac{4}{3}\pi r^3 \\
T/[\mathrm{K}] &= T/[^\circ\mathrm{C}] + 273.15 \\
T/[^\circ\mathrm{C}] &= (T/[^\circ\mathrm{F}] - 32) \cdot \frac{5}{9} \\
F(r) &= \frac{q_1 q_2}{4\pi\varepsilon_0 r^2} \\
V(r) &= \frac{q_1 q_2}{4\pi\varepsilon_0 r} \\
V_\mathrm{eff} &= -\frac{Z_\mathrm{eff} e^2}{4\pi\varepsilon_0 r} \\
T &= \frac{1}{2}mv^2 \\
p &= mv \\
L &= mvr \\
T &= -\frac{V}{2} \\
\hat{\mathcal{H}} &= \hat{\mathcal{T}} + \hat{\mathcal{V}} \\
\nu &= \frac{c}{\lambda} \\
\vec{E} &= \vec{E}_0 \sin\left\{\vec{k} \cdot \vec{z} + \omega t + \Phi\right\} \\
\left|\vec{k}\right| &= \frac{1}{\lambda} \\
\omega &= 2\pi\nu \\
2\ell &= n\lambda \\
E_n &= nh\nu \\
E_\mathrm{photon} &= h\nu = \frac{hc}{\lambda} \\
L_n &= n\hbar = n\frac{h}{2\pi} \\
E_n &= -\frac{hcR_\mathrm{H}}{n^2} \\
h\nu &= E_{n_2} - E_{n_1} \\
\nu\left(n_2 \longrightarrow n_1\right) &= cR_\mathrm{H}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) \\
\bar{\nu}\left(n_2 \longrightarrow n_1\right) &= R_\mathrm{H}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) \\
\lambda\left(n_2 \longrightarrow n_1\right) &= \frac{1}{R_\mathrm{H}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)} \\
R_\infty &= \frac{m_e e^4}{8\varepsilon_0^2 h^3 c} \\
R_\mathrm{H} &= \frac{\mu_H e^4}{8\varepsilon_0^2 h^3 c} \\
\mu &= \frac{m_1 \cdot m_2}{m_1 + m_2} \\
\lambda &= \frac{h}{p} \\
n\lambda &= 2\pi \cdot r \\
\Delta x \cdot \Delta p_x &\leq \frac{\hbar}{2} \\
\Delta\phi \cdot \Delta L &\leq \frac{\hbar}{2} \\
\Delta E \cdot \Delta t &\leq \frac{\hbar}{2}
\end{aligned}
$$

$$\hat{\mathcal{H}} \cdot \psi(x,y,z) = E \cdot \psi(x,y,z)$$
$$\hat{\mathcal{H}} \cdot \psi(r,\theta,\phi) = E \cdot \psi(r,\theta,\phi)$$
$$x = r\sin\theta\cos\phi$$
$$y = r\sin\theta\sin\phi$$
$$z = r\cos\theta$$
$$\psi(r,\theta,\phi) = R_{nl}(r) \cdot \Theta_{lm_l}(\theta) \cdot \Phi_{m_l}(\phi)$$
$$n = 1, 2, \ldots$$
$$l = 0, 1, 2, \ldots, n-1$$
$$m_l = 0, \pm 1, \pm 2, \ldots, \pm l$$
$$m_s = \pm\frac{1}{2}$$
$$E_n = \frac{-hcR_{\mathrm{H}}}{n^2} = -\frac{\mu e^4}{8\varepsilon_0^2 h^2 n^2}$$
$$R_\infty = \frac{m_e e^4}{8\varepsilon_0^2 h^3 c}$$
$$R_{\mathrm{H}} = \frac{\mu e^4}{8\varepsilon_0^2 h^3 c}$$
$$L = I \cdot \omega$$
$$\left|\vec{L}\right| = \sqrt{l(l+1)}\,\hbar$$
$$\left|\vec{S}\right| = \sqrt{s(s+1)}\,\hbar$$
$$\vec{\mu} = \frac{-g_e e}{2m_e}\vec{S}$$
$$\left|\vec{L}_z\right| = |m_l \cdot \hbar|$$
$$\left|\vec{S}_z\right| = |m_s \cdot \hbar|$$
$$L_z = \left|\sum_i m_l(i)\right|$$
$$S = \left|\sum_i m_s(i)\right|$$
$$\mathbf{mole{-}\%}(i) = \frac{n_i}{\sum\limits_i n_i}$$
$$\mathbf{mass{-}\%}(i) = \frac{m_i}{\sum\limits_i m_i}$$
$$\overline{\mathcal{M}} = \sum_i c_i \cdot \mathcal{M}_i$$
$$1 = \sum_i c_i$$
$$n = \frac{m}{\mathcal{M}}$$
$$\text{bond order} = \frac{1}{2}\left(\begin{array}{c}\text{\# of } e^-\text{'s in}\\ \text{bonding MOs}\end{array} - \begin{array}{c}\text{\# of } e^-\text{'s in}\\ \text{antibonding MOs}\end{array}\right)$$
$$\text{bond order} = \left(\begin{array}{c}\text{average \# of bonds}\\ \text{averaged over all}\\ \text{resonance structures}\end{array}\right)$$

$$d = \frac{m}{V}$$
$$n = c_{\mathrm{init}} \cdot V_{\mathrm{init}} = c_{\mathrm{fin}} \cdot V_{\mathrm{fin}}$$
$$M = \frac{n_{\mathrm{solute}}}{V_{\mathrm{solution}}}$$
$$b = \frac{n_{\mathrm{solute}}}{m_{\mathrm{solvent}}}$$
$$\chi = \frac{n_{\mathrm{solute}}}{n_{\mathrm{total}}}$$
$$M \approx d_{\mathrm{solvent}} \cdot b$$
$$n_{\mathrm{ions}} = i \cdot n_{\mathrm{formula\ unit}}$$
$$\%\text{-yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \cdot 100\%$$
$$PV = nRT$$
$$P_{\mathrm{total}} = \sum_i P_i$$
$$Z = \frac{PV}{nRT}$$
$$w = F \cdot d$$
$$F = \frac{P}{A}$$
$$E_{\mathrm{grav}} = mg\Delta h$$
$$E_{\mathrm{kin}} = \frac{m}{2}v^2$$
$$\langle E_{\mathrm{kin}}\rangle = \frac{3}{2}k_{\mathrm{B}}T$$
$$v_{\mathrm{mp}} = \sqrt{\frac{2\,R\,T}{\mathcal{M}}}$$
$$\langle v\rangle = \sqrt{\frac{8\,R\,T}{\pi\mathcal{M}}}$$
$$v_{\mathrm{rms}} = \sqrt{\frac{3\,R\,T}{\mathcal{M}}}$$
$$w = -P_{\mathrm{ext}}\,\Delta V$$
$$q = C \cdot \Delta T$$
$$C = c \cdot m = \overline{C} \cdot n$$
$$\Delta U = q + w$$
$$H = U + PV$$
$$\Delta H = \Delta U + \Delta(PV)$$
$$U = \frac{3}{2}nRT$$
$$\Delta_{\mathrm{rxn}}\overline{H}^\circ = \sum_{\mathrm{products}} \Delta_{\mathrm{f}}\overline{H}^\circ - \sum_{\mathrm{reactants}} \Delta_{\mathrm{f}}\overline{H}^\circ$$

Considerations for Z_{eff} and E_i calculations (Slater's Rules):

$$E_i \approx (-1312\ \mathrm{kJ/mol})\left(\frac{Z_{\mathrm{eff}}(i)}{n_i}\right)^2$$

group	other electrons in same group	electrons in group w/ same n and $< l$	electrons in group w/ $n-1$	electrons in group $< (n-1)$
$[1s]$	0.30	-	-	-
$[ns, np]$	0.35	-	0.85	1
$[nd]$ or $[nf]$	0.35	1	1	1

Pauling Electronegativities

H 2.20																	He –
Li 0.98	Be 1.57											B 2.04	C 2.55	N 3.04	O 3.44	F 3.98	Ne –
Na 0.93	Mg 1.31											Al 1.61	Si 1.90	P 2.19	S 2.58	Cl 3.16	Ar –
K 0.82	Ca 1.0	Sc 1.36	Ti 1.54	V 1.63	Cr 1.66	Mn 1.55	Fe 1.83	Co 1.88	Ni 1.91	Cu 1.90	Zn 1.65	Ga 1.81	Ge 2.01	As 2.18	Se 2.55	Br 2.96	Kr 3.0
Rb 0.82	Sr 0.95	Y 1.22	Zr 1.33	Nb 1.6	Mo 2.16	Tc 1.9	Ru 2.2	Rh 2.28	Pd 2.20	Ag 1.93	Cd 1.69	In 1.78	Sn 1.96	Sb 2.05	Te 2.1	I 2.66	Xe 2.6
Cs 0.79	Ba 0.89	Lu 1.27	Hf 1.3	Ta 1.5	W 2.36	Re 1.9	Os 2.2	Ir 2.20	Pt 2.28	Au 2.54	Hg 2.0	Tl 1.62	Pb 2.33	Bi 2.02	Po 2.0	At 2.2	Rn –
Fr >0.79	Ra 0.9	Lr –	Rf –	Db –	Sg –	Bh –	Hs –	Mt –	Ds –	Rg –	Cn –	Nh –	Fl –	Mc –	Lv –	Ts –	Og –

La 1.10	Ce 1.12	Pr 1.13	Nd 1.14	Pm –	Sm 1.17	Eu –	Gd 1.20	Tb –	Dy 1.22	Ho 1.23	Er 1.24	Tm 1.25	Yb –
Ac 1.1	Th 1.3	Pa 1.5	U 1.38	Np 1.36	Pu 1.28	Am 1.3	Cm 1.3	Bk 1.3	Cf 1.3	Es 1.3	Fm 1.3	Md 1.3	No 1.3

<u>Selected Brønsted-Lowry Acids and Bases:</u>
Strong Acids: $H_2S_2O_7$, HI, HBr, HCl, H_2SO_4, $HClO_4$, HNO_3, $H_3O^+(aq)$.
Weak Acids: $HSO_4^-(aq)$, $H_2C_2O_4(aq)$, $H_2CrO_4(aq)$, $Co(OH_2)_6^{3+}(aq)$, $H_3PO_4(aq)$, $Fe(OH_2)_6^{3+}(aq)$, $H_3AsO_4(aq)$, HF(aq), $HC_2O_4^-(aq)$, $CH_3COOH(aq)$, $Al(OH_2)_6^{3+}(aq)$, $H_2CO_3(aq)$.,$NH_4^+(aq)$, $H_2S(aq)$ HCN(aq).
Strong Bases: O^{2-}, S^{2-}, NH_2^-, $OH^-(aq)$.
Weak Bases: $PO_4^{3-}(aq)$, $CO_3^{2-}(aq)$, $NH_3(aq)$, $CN^-(aq)$, $HPO_4^{2-}(aq)$, $HS^-(aq)$, $HAsO_4^{2-}(aq)$, $CrO_4^{2-}(aq)$, $HCO_3^{2-}(aq)$, $CH_3COO^-(aq)$, $C_2O_4^{2-}(aq)$, $F^-(aq)$, $SO_4^{2-}(aq)$.

<u>Selected Polyatomic Ions:</u>

Name	Formula	Name	Formula	Name	Formula
ammonium	NH_4^+	bicarbonate	HCO_3^-	phosphate	PO_4^{3-}
hydronium	H_3O^+	nitrate	NO_3^-	hydrogen phosphate	HPO_4^{2-}
peroxide	O_2^{2-}	nitrite	NO_2^-	dihydrogen phosphate	$H_2PO_4^-$
hydroxide	OH^-	sulfate	SO_4^{2-}	perchlorate	ClO_4^-
acetate	CH_3COO^-	bisulfate	HSO_4^-	chlorate	ClO_3^-
cyanide	CN^-	sulfite	SO_3^{2-}	chlorite	ClO_2^-
azide	N_3^-	bisulfite	HSO_3^-	hypochlorite	ClO^-
carbonate	CO_3^{2-}	hydrogen sulfide	HS^-	permanganate	MnO_4^-

IUPAC Periodic Table of the Elements

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1 H hydrogen 1.008 [1.0078, 1.0082]																	2 He helium 4.0026
3 Li lithium 6.94 [6.938, 6.997]	4 Be beryllium 9.0122											5 B boron 10.81 [10.806, 10.821]	6 C carbon 12.011 [12.009, 12.012]	7 N nitrogen 14.007 [14.006, 14.008]	8 O oxygen 15.999 [15.999, 16.000]	9 F fluorine 18.998	10 Ne neon 20.180
11 Na sodium 22.990	12 Mg magnesium 24.305 [24.304, 24.307]											13 Al aluminium 26.982	14 Si silicon 28.085 [28.084, 28.086]	15 P phosphorus 30.974	16 S sulfur 32.06 [32.059, 32.076]	17 Cl chlorine 35.45 [35.446, 35.457]	18 Ar argon 39.95 [39.792, 39.963]
19 K potassium 39.098	20 Ca calcium 40.078(4)	21 Sc scandium 44.956	22 Ti titanium 47.867	23 V vanadium 50.942	24 Cr chromium 51.996	25 Mn manganese 54.938	26 Fe iron 55.845(2)	27 Co cobalt 58.933	28 Ni nickel 58.693	29 Cu copper 63.546(3)	30 Zn zinc 65.38(2)	31 Ga gallium 69.723	32 Ge germanium 72.630(8)	33 As arsenic 74.922	34 Se selenium 78.971(8)	35 Br bromine 79.904 [79.901, 79.907]	36 Kr krypton 83.798(2)
37 Rb rubidium 85.468	38 Sr strontium 87.62	39 Y yttrium 88.906	40 Zr zirconium 91.224(2)	41 Nb niobium 92.906	42 Mo molybdenum 95.95	43 Tc technetium	44 Ru ruthenium 101.07(2)	45 Rh rhodium 102.91	46 Pd palladium 106.42	47 Ag silver 107.87	48 Cd cadmium 112.41	49 In indium 114.82	50 Sn tin 118.71	51 Sb antimony 121.76	52 Te tellurium 127.60(3)	53 I iodine 126.90	54 Xe xenon 131.29
55 Cs caesium 132.91	56 Ba barium 137.33	57-71 lanthanoids	72 Hf hafnium 178.49(2)	73 Ta tantalum 180.95	74 W tungsten 183.84	75 Re rhenium 186.21	76 Os osmium 190.23(3)	77 Ir iridium 192.22	78 Pt platinum 195.08	79 Au gold 196.97	80 Hg mercury 200.59	81 Tl thallium 204.38 [204.38, 204.39]	82 Pb lead 207.2	83 Bi bismuth 208.98	84 Po polonium	85 At astatine	86 Rn radon
87 Fr francium	88 Ra radium	89-103 actinoids	104 Rf rutherfordium	105 Db dubnium	106 Sg seaborgium	107 Bh bohrium	108 Hs hassium	109 Mt meitnerium	110 Ds darmstadtium	111 Rg roentgenium	112 Cn copernicium	113 Nh nihonium	114 Fl flerovium	115 Mc moscovium	116 Lv livermorium	117 Ts tennessine	118 Og oganesson

Key:

atomic number
Symbol
name
conventional atomic weight
standard atomic weight

57 La lanthanum 138.91	58 Ce cerium 140.12	59 Pr praseodymium 140.91	60 Nd neodymium 144.24	61 Pm promethium	62 Sm samarium 150.36(2)	63 Eu europium 151.96	64 Gd gadolinium 157.25(3)	65 Tb terbium 158.93	66 Dy dysprosium 162.50	67 Ho holmium 164.93	68 Er erbium 167.26	69 Tm thulium 168.93	70 Yb ytterbium 173.05	71 Lu lutetium 174.97
89 Ac actinium	90 Th thorium 232.04	91 Pa protactinium 231.04	92 U uranium 238.03	93 Np neptunium	94 Pu plutonium	95 Am americium	96 Cm curium	97 Bk berkelium	98 Cf californium	99 Es einsteinium	100 Fm fermium	101 Md mendelevium	102 No nobelium	103 Lr lawrencium

INTERNATIONAL UNION OF PURE AND APPLIED CHEMISTRY

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2019 IYPT International Year of the Periodic Table of Chemical Elements

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